

Mastercard AI Defence Lab

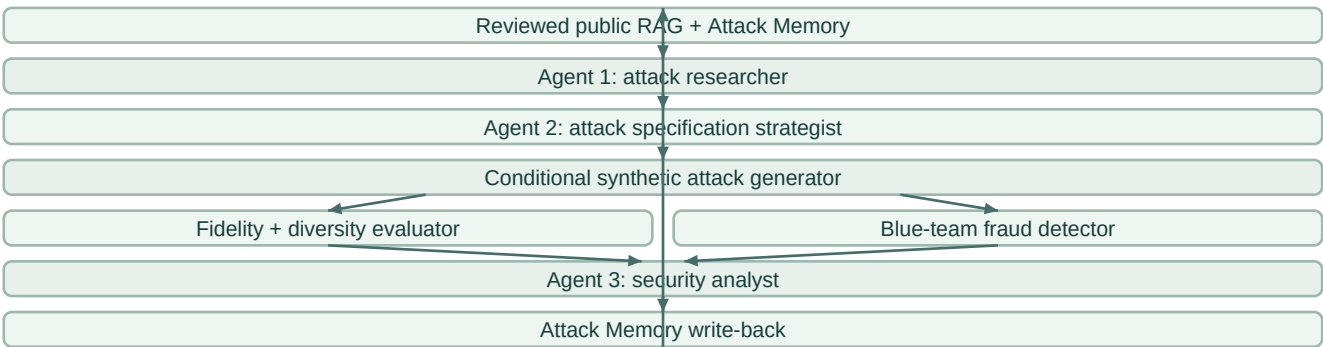
Current implementation state report | Generated: 2026-08-25

Current State

The project is an offline synthetic-only red-team/blue-team prototype. The closed loop is operational with a three-round protocol, unseen-attack evaluation, and memory-guided adaptation.

- Data scope is synthetic only; no real cardholder, PII, production, or live-system data is used.
- Execution supports local orchestration and Kaggle GPU mode for heavy model and generator workloads.
- A shared Qwen2.5 model is used for three roles: Agent 1 research, Agent 2 specification, Agent 3 security analysis.
- The feedback path is intentionally constrained: Agent 3 findings are persisted to Attack Memory and then used by Agent 1 in the next round.

Architecture



Architecture loop with enforced feedback path: Agent 3 -> Memory -> Agent 1

Flow Blocks Explained

- Reviewed public RAG + Attack Memory: retrieval context that mixes approved public summaries with recent weakness history.
- Agent 1: proposes the next attack direction as a structured AttackHypothesis.
- Agent 2: converts the hypothesis into a structured AttackSpecification with feature constraints.
- Conditional synthetic attack generator: creates synthetic fraud rows by attack family profile.
- Fidelity + diversity evaluator: checks realism-like similarity and variation of generated attacks before interpretation.
- Blue-team fraud detector: scores fraud probability and computes precision/recall/F1/ROC-AUC/FPR.
- Agent 3: analyzes weaknesses from detector and fidelity evidence, then writes recommendations.
- Attack Memory write-back: stores hypothesis/specification/evaluation/weakness records for next-round adaptation.

Artefact To File Map

- Knowledge base manifest and provenance: data/knowledge_base/manifest.json
- Knowledge base usage policy and safety constraints: data/knowledge_base/README.md
- Seed domain knowledge used in retrieval: data/knowledge_base/seed_fraud_typologies.txt

- Reviewed-source defensive summaries used for retrieval: data/knowledge_base/sources/*.txt
- Knowledge retrieval implementation: src/mastercard_defence/rag.py
- Synthetic attack generation schema and family profiles: src/mastercard_defence/synthetic.py
- Adaptive attack-family run records (round-level samples): adaptive/adaptive_qwen_ctgan_results_20260824T105805Z.json
- Attack family taxonomy data: adaptive/taxonomy_v1.json
- Attack memory database written each round: artifacts/attack_memory.sqlite

Knowledge Base Samples

These are direct excerpts from the data files used by LocalKnowledgeBase retrieval for Agent 1 attack research.

Sample KB excerpt 1

Public seed knowledge for the first local demonstration.

Payment fraud can combine social engineering, account takeover, trusted-device abuse, beneficiary manipulation, and low-and-slow transaction patterns. Defenses should compare amount, time, device, beneficiary, channel, and short-window velocity signals. This seed is a synthetic planning aid

Sample KB excerpt 2

Source summary: NIST AI 600-1 Generative Artificial Intelligence Profile

NIST's Generative AI Profile extends AI risk-management guidance to risks that are distinctive or amplified in generative AI systems. For this project, it supports testing model reliability, documenting generated-content limitations, monitoring misuse risk, and retaining trac

Attack Data Samples

The current repository stores attack records primarily as round-level JSON evidence and taxonomy entries. Representative examples are shown below.

Attack sample shape required by detector includes features amount, hour, device_change, beneficiary_change, velocity_24h, channel and label is_fraud.

family_selection_mode meaning: seeded_plan = fixed initial schedule, agent_1_adaptive_recommendation = selected by Agent 1 using prior weakness and memory.

Row-level generation sample could not be produced at build time.

Round Metrics Record Samples

Sample attack-related record 1

```
{
  "seed": 20260821,
  "round": 1,
  "attack_family": "account_takeover",
  "family_selection_mode": "seeded_plan",
  "recall": 0.6625,
  "f1": 0.7969924812030075,
  "novelty_score": 1.0
}
```

Sample attack-related record 2

```
{
  "seed": 20260821,
  "round": 2,
  "attack_family": "trusted_device",
  "family_selection_mode": "agent_1_adaptive_recommendation",
  "recall": 0.6875,
  "f1": 0.8088235294117647,
  "novelty_score": 0.8758
}
```

Sample attack-related record 3

```
{
  "seed": 20260821,
  "round": 3,
  "attack_family": "beneficiary_manipulation",
  "family_selection_mode": "agent_1_adaptive_recommendation",
  "recall": 0.7,
  "f1": 0.8175182481751825,
  "novelty_score": 0.96
}
```

```
}
```

Sample attack-related record 4

```
{
  "seed": 20260821,
  "round": 4,
  "attack_family": "low_and_slow",
  "family_selection_mode": "agent_1_adaptive_recommendation",
  "recall": 0.625,
  "f1": 0.7633587786259542,
  "novelty_score": 0.9609
}
```

Taxonomy Record Samples

These records define approved attack families and observable feature expectations used by the benchmark taxonomy.

Sample taxonomy record 1

```
{
  "name": "account_takeover",
  "definition": "Synthetic payment behavior following account compromise.",
  "observable_features": [
    "device_change",
    "beneficiary_change",
    "velocity_24h",
    "amount"
  ]
}
```

Sample taxonomy record 2

```
{
  "name": "trusted_device",
  "definition": "Synthetic fraudulent behavior that preserves familiar-device signals.",
  "observable_features": [
    "device_change",
    "beneficiary_change",
    "velocity_24h"
  ]
}
```

Blue-Team Detector

Model implementation is scikit-learn Pipeline: ColumnTransformer with OneHotEncoder(handle_unknown='ignore') for channel, followed by HistGradientBoostingClassifier(random_state=42).

Training/evaluation input schema: [amount, hour, device_change, beneficiary_change, velocity_24h, channel] and target label is_fraud in {0,1}.

Metric Formulas

Precision = $TP / (TP + FP)$, Recall = $TP / (TP + FN)$, $F1 = 2 * (Precision * Recall) / (Precision + Recall)$.

False Positive Rate = $FP / (FP + TN)$. ROC-AUC is the area under the ROC curve formed by TPR versus FPR across score thresholds.

Fidelity And Diversity Evaluator

Fidelity computes amount_mean_delta, amount_std_delta, behavioural_signal_delta, and channel_coverage comparing unseen attacks to synthetic legitimate reference.

Behavioural plausibility formula: $plausibility = \max(0, 1 - \min(1, 0.25 * (\text{mean_delta} / \text{ref_mean}) + 0.25 * (\text{std_delta} / \text{ref_std}) + 0.25 * \text{behaviour_delta} + 0.25 * (1 - \text{channel_coverage})))$.

Diversity includes unique_row_ratio = unique_rows / total_rows, family_coverage_ratio = attack_family_count / 7, and normalized channel entropy $H = -\sum(p_i * \log_2(p_i)) / \log_2(3)$.

Detector Input Sample Table

The table below shows 10 sample input rows (5 legitimate + 5 attack) with the exact feature columns and label consumed by the blue-team detector.

attack_family	amount	hour	device	beneficiary	velocity_24h	channel	is_fraud
legitimate	42.50	9	0	0	2	mobile	0
legitimate	27.19	13	0	0	1	web	0

attack_family	amount	hour	device	beneficiary	velocity_24h	channel	is_fraud
legitimate	65.22	16	0	0	3	mobile	0
legitimate	18.74	21	0	0	1	card_present	0
legitimate	54.01	8	0	0	2	web	0
trusted_device	112.43	19	0	1	5	mobile	1
trusted_device	87.65	22	0	1	4	web	1
trusted_device	143.20	18	0	0	6	mobile	1
trusted_device	99.87	20	0	1	5	card_present	1
trusted_device	121.56	23	0	1	4	web	1

Implemented Components

- Local knowledge base retrieval over reviewed public summaries for evidence grounding.
- SQLite Attack Memory that stores hypotheses, specifications, evaluations, and weakness reports.
- Conditional synthetic attack generation with family constraints and deterministic fallback support.
- Blue-team detector based on scikit-learn with unseen synthetic attack evaluation against legitimate holdout.
- Fidelity, diversity, and novelty indicators recorded each round as internal experiment evidence.
- Robustness suite support for multi-seed runs and aggregate metric summaries.

Status and Gaps

The implementation is strong as an engineering baseline and has reproducible loop behavior. Remaining work is primarily submission hardening: calibration, expanded per-family analysis, UI polish, and final evidence packaging.

Recommendation

Use this report as the single current-state reference artifact in the workspace. Keep future PDF outputs versioned and date-stamped to avoid stale report accumulation.